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1 Introduction

Low Earth Orbit (LEO) satellite networks are characterized by highly dynamic topologies due to the predictable orbital motion of satellites. Classical shortest-path routing algorithms such as Dijkstra typically rely on static edge weights (e.g., propagation delay), and therefore do not explicitly account for satellite mobility.

In this project, we extend the `Hypatia+` simulation framework to implement and evaluate a **direction-aware routing algorithm** that biases routing decisions based on the relative motion of satellites. The objective is to evaluate whether incorporating mobility information into routing decisions can improve performance compared to baseline routing algorithms.

Objectives

- Implement a mobility-aware extension of Dijkstra's algorithm
- Compare its performance against standard Dijkstra and Frog routing
- Analyze the impact of directional bias on routing performance

2 System Model

2.1 LEO Network Model

The simulations are conducted using the `Hypatia+` framework. The network consists of:

- A Walker-type LEO satellite constellation
- Inter-satellite links (ISLs) between neighboring satellites
- Ground stations acting as traffic sources and destinations

The satellite positions and velocities evolve deterministically according to orbital mechanics. At each simulation snapshot, the network is represented as a directed graph:

$$G = (V, E)$$

where V is the set of satellites and E is the set of available ISLs.

2.2 Baseline Routing Algorithms

We evaluate the following routing schemes:

- **Standard Dijkstra**: shortest-path routing using static edge weights (e.g., propagation delay)
- **Frog Routing**: a mobility-aware routing algorithm provided in the framework
- **Direction-Aware Dijkstra**: the algorithm implemented in this project

3 Direction-Aware Routing Algorithm

3.1 Directional Alignment Metric

For each satellite, define:

- $\mathbf{v}$: velocity vector of the transmitting satellite
- $\mathbf{d}$: direction vector of the inter-satellite link

For a directed link from satellite i to satellite j, define the link direction vector:

$$\mathbf{d}_{ij} = \frac{\mathbf{p}_j - \mathbf{p}_i}{\|\mathbf{p}_j - \mathbf{p}_i\|}$$

where $\mathbf{p}_i$ and $\mathbf{p}_j$ are satellite positions.

The directional alignment between motion and transmission is computed as:

$$\alpha = \cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{d}}{\|\mathbf{v}\|\|\mathbf{d}\|}$$

where $\alpha \in [-1, 1]$.

3.2 Edge Weight Modification

Let w_0 be the baseline edge weight (e.g., propagation delay). The modified edge weight is defined as:

$$w = w_0 \cdot (1 + \beta \cdot f(\alpha))$$

where:

- β is a tunable parameter controlling the strength of the directional bias
- $f(\alpha)$ is a monotonic function of α

A positive alignment (signal forwarded in the direction of motion) results in a lower effective cost, while a negative alignment increases the cost.

3.3 Routing Procedure

At each network snapshot:

1. Compute directional alignment for all ISLs
2. Modify edge weights accordingly
3. Run Dijkstra's algorithm using the modified weights
4. Record the selected paths and performance metrics

4 Experimental Methodology

4.1 Simulation Scenarios

We evaluate routing performance under multiple configurations:

- Different values of the directional bias parameter β
- Multiple random source–destination pairs for statistical confidence
- Ground-to-ground communication via satellites
- Different constellation sizes or orbital parameters

Each configuration should be repeated sufficiently to obtain statistically meaningful results.

4.2 Performance Metrics

The following metrics must be reported:

- End-to-end latency
- Packet delivery ratio
- Average hop count
- Path stability (path changes over time)

Results should be presented using averages and confidence intervals, and compared across all routing algorithms.

4.3 Expected Outcomes

Students are expected to:

- Identify scenarios where direction-aware routing outperforms standard Dijkstra
- Analyze sensitivity to the parameter β
- Discuss trade-offs between latency reduction and path stability
- When and why mobility-aware routing outperforms standard Dijkstra
- Scenarios where Frog routing performs better
- Trade-offs between latency optimization and route stability

The discussion should emphasize *why* observed trends occur, not only numerical improvements.

5 Expected Deliverables

Students are expected to submit:

- Modified routing code
- Experiment configuration files
- Plots and tables summarizing results
- A short written report interpreting the findings

6 Conclusion

This project demonstrates how predictable satellite mobility can be leveraged to improve routing decisions in LEO networks. We critically assess both the benefits and limitations of incorporating direction-awareness into shortest-path routing.